

Nephrology Clinical Assessment Report

Chronic Kidney Disease Screening · Confidential Clinical Document

Patient Information

Patient Name	Margaret T.	Patient ID	CKD-2026-002
Date of Report	April 29, 2026	Exam Type	Nephrology Screening — CKD Assessment
Age / Gender	65 years · Female	Referring Department	Internal Medicine
Hypertension	Yes — currently managed	Diabetes Mellitus	Yes — confirmed
Coronary Artery Disease	Yes — confirmed	Informant	Patient / Medical Record

Laboratory Findings — Quantitative Markers

Parameter	Result	Reference Range	Status	Clinical Note
Blood Pressure (bp)	160 mmHg	<120 mmHg	HIGH	Stage 2 hypertension — accelerates glomerular damage
Blood Sugar (bgr)	Positive	Negative	ABNORMAL	Glycosuria — poorly controlled diabetes
Blood Urea (bu)	90 mg/dL	7–20 mg/dL	VERY HIGH	Severe azotemia — marked reduction in renal clearance
Serum Creatinine (sc)	3.5 mg/dL	0.5–1.1 mg/dL	VERY HIGH	eGFR critically reduced — CKD Stage 4–5
Sodium (sod)	132 mEq/L	136–145 mEq/L	LOW	Hyponatremia — impaired tubular sodium reabsorption
Potassium (pot)	6.5 mEq/L	3.5–5.0 mEq/L	HIGH	Hyperkalemia — risk of fatal cardiac arrhythmia
Hemoglobin (hemo)	8.0 g/dL	12.0–16.0 g/dL	VERY LOW	Severe renal anemia — reduced erythropoietin production
Packed Cell Volume (pcv)	25%	36–48%	LOW	Hematocrit critically reduced — confirms anemia severity
WBC Count (wc)	Abnormal	Normal	ABNORMAL	Elevated urinary WBCs — active renal inflammation
RBC Count (rc)	Abnormal	Normal	ABNORMAL	Dysmorphic RBCs — glomerulonephritis or tubular bleeding

Specific Gravity (sg)	1.010	1.010–1.030	VERY LOW	Impaired urinary concentration — tubular dysfunction
Albumin (al)	2+	Negative / Trace	HIGH	Significant proteinuria — glomerular membrane compromise

Laboratory Findings — Qualitative & Clinical Markers

Marker	Finding	Status	Clinical Significance
RBC in Urine (rbc)	Abnormal	FLAGGED	Hematuria — glomerular or tubular bleeding
Pus Cells (pc)	Abnormal	FLAGGED	Pyuria — active UTI or tubulointerstitial nephritis
Pus Cell Clumps (pcc)	Present	FLAGGED	Strong indicator of upper UTI or renal abscess
Bacteria (ba)	Present	FLAGGED	Confirmed bacteriuria — active urinary tract infection
Diabetes Mellitus (dm)	Yes	FLAGGED	Leading cause of CKD — diabetic nephropathy likely etiology
Coronary Artery Disease (cad)	Yes	FLAGGED	Cardiovascular complication — worsens CKD prognosis bidirectionally
Appetite (appet)	Poor	FLAGGED	Uremic anorexia — hallmark symptom of advanced renal failure
Pedal Edema (pe)	Present	FLAGGED	Bilateral fluid retention — hallmark of nephrotic syndrome / CKD
Anemia (ane)	Yes	FLAGGED	Renal anemia from EPO deficiency — clinically severe at Hb 8.0

Model Input Features

bp	160	pot	6.5
su	2	wc	abnormal
rbc	abnormal	rc	abnormal
pc	abnormal	dm	yes
pcc	present	cad	yes
ba	present	appet	poor
bgr	positive	pe	yes
bu	90	ane	yes
sod	132		

These feature values are extracted directly from clinical findings and formatted for ingestion by the CKD prediction model.

Clinical Observations

Margaret T. is a 65-year-old female with a confirmed triad of hypertension, diabetes mellitus, and coronary artery disease — representing three of the most potent independent risk factors for chronic kidney disease. She presents with poor appetite, bilateral pedal edema, and confirmed clinical anemia. Urinalysis reveals significant proteinuria (2+), glycosuria, bacteriuria, pyuria, and the presence of pus cell clumps, indicating an active urinary tract infection superimposed on advanced structural renal disease. Both RBC and WBC classifications are abnormal. Serum markers confirm severe azotemia (BU 90 mg/dL), critically elevated creatinine (3.5 mg/dL), hyperkalemia (K⁺ 6.5 mEq/L), and hyponatremia (Na⁺ 132 mEq/L). Hematological indices confirm severe anemia with hemoglobin at 8.0 g/dL and PCV at 25%.

Clinical Impression

The clinical, laboratory, and symptomatic profile of Margaret T. is consistent with end-stage Chronic Kidney Disease (CKD Stage 4–5), most likely secondary to diabetic nephropathy and hypertensive nephrosclerosis. The total convergence of all 17 model input features in the pathological direction — spanning renal function, urinalysis, hematology, electrolyte balance, active infection, and systemic comorbidities — constitutes one of the most complete and severe CKD presentations. The elevated potassium (6.5 mEq/L) poses an immediate risk of life-threatening cardiac arrhythmia and requires urgent intervention. This report does not substitute for a formal clinical diagnosis.

Recommendations

- Emergency electrolyte management:** Hyperkalemia (K⁺ 6.5) requires immediate treatment — calcium gluconate, insulin-dextrose, or kayexalate; continuous cardiac monitoring.
- Urgent nephrology referral for formal CKD staging and renal replacement therapy planning.
- Antibiotic therapy:** Treat confirmed UTI with culture-guided antibiotics; upper UTI protocol given pus cell clumps and bacteriuria.
- ESA therapy and iron status evaluation for renal anemia (Hb 8.0 g/dL).
- Optimize antihypertensive and glycemic regimens — target BP <130/80 mmHg; ACE inhibitor or ARB preferred for renoprotection in diabetic nephropathy.
- Dietary nephrology consultation: potassium and phosphate restriction, fluid balance management.
- Cardiology co-management given confirmed CAD and severe anemia.

Examining Clinician — Nephrology

Date: April 29, 2026